Gowtham Alasakani

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SUMMARY

Data Engineer and former Data Engineer/Data Analyst with experience building scalable data pipelines, ETL/ELT workflows, cloud data platforms, analytics solutions, and machine learning applications. Skilled in Python, SQL, PySpark, AWS, Snowflake, Databricks, Dataiku, dbt, Airbyte, and workflow orchestration, with a strong focus on data integration, performance optimization, data quality, validation, reconciliation, governance, monitoring, and production support. Developed analytics-ready data products, automated reporting and validation workflows, and machine learning and computer vision solutions using TensorFlow, OpenCV, Pandas, and NumPy across healthcare, pharmaceutical, and operational data environments. Ready to apply this end-to-end experience to deliver reliable data solutions, strengthen business intelligence, and support faster, more informed decision-making.

SKILLS

- **Programming & Data Analysis**: Python, SQL, PySpark, SparkSQL, Pandas, NumPy, Jupyter Notebook, Jupyter, Scikit-learn, SQLAlchemy, Exploratory Data Analysis, Statistical Analysis, Data Profiling, Trend Analysis, Root-Cause Analysis, Exception Analysis, Feature Engineering
- **Data Engineering & Data Platforms**: ETL, ELT, Data Engineering, Data Pipelines, Data Integration, Data Ingestion, Data Processing, Data Transformation, Data Modeling, Dataiku, Snowflake, Snowpark, Databricks, dbt, Airbyte, Data Warehouse, Analytics Engineering, Apache Spark
- **Cloud, DevOps & Infrastructure**: AWS, Azure, Amazon S3, Amazon EKS, AWS Lambda, AWS IAM, AWS Secrets Manager, Amazon CloudWatch, Amazon SQS, Amazon SNS, Amazon EventBridge, AWS Step Functions, AWS CodePipeline, AWS CodeBuild, AWS CodeDeploy, Terraform, Docker, Git, GitHub Actions, Jenkins, CI/CD, DevOps
- **Databases, SQL & Reporting**: Snowflake, MySQL, PostgreSQL, SQL Server, MS SQL Server, Oracle, Advanced SQL, Complex SQL Queries, Analytical SQL, Query Optimization, CTEs, Window Functions, Stored Procedures, Excel, Advanced Excel, Pivot Tables, Power Query, Power BI, Dashboard Development, Business Intelligence, Data Visualization, KPI Reporting, Executive Reporting
- **Data Quality, Governance & Operations**: Data Quality, Data Quality Management, Data Validation, Data Reconciliation, Data Integrity, Data Governance, Data Lineage, Data Catalog, Metadata Management, Data Governance Standards, Quality Assurance, SOP Compliance, Process Testing, UAT, Monitoring, Production Support, Incident Management, SLA Management, Operational Risk, Process Optimization, Workflow Analysis, Operational Reporting, Governance, Data Security, Least Privilege Access
- **Machine Learning, Computer Vision & Delivery**: Machine Learning, Supervised Learning, Convolutional Neural Networks, TensorFlow, OpenCV, Image Processing, Image Preprocessing, Image Augmentation, Model Training, Model Evaluation, Hyperparameter Tuning, Model Validation, Classification, Accuracy Analysis, Confusion Matrix, Stakeholder Management, Cross-Functional Collaboration, Client Communication, Requirements Gathering, Requirements Analysis, Solution Design, Technical Documentation, SOP Documentation, Agile, Scrum, Jira, Project Tracking, Risk Management, Status Reporting, Onshore and Offshore Coordination, Team Leadership, Healthcare Analytics, Healthcare Operations, Pharmaceutical Data, Regulated Data Environments, Secondary Research

EXPERIENCE

Astir IT Solution's **Oct 2025 - Present**

AWS Data Engineer *Naperville | IL*

- Designed, built, and operated AWS data engineering solutions with Python, PySpark, SparkSQL, Amazon S3, Databricks, Snowflake, dbt, and Airbyte, developing scalable ingestion and transformation pipelines that integrated APIs, databases, files, and enterprise source systems into analytics-ready data platforms.
- Developed and maintained Prefect workflows with reusable flows and tasks, scheduled and parameterized deployments, retry logic, failure handling, and recovery strategies monitored production runs through Prefect UI and Cloud to troubleshoot failed workflows and maintain reliable data delivery.
- Built data quality and validation processes using Python, PySpark, SQL, and dbt tests to detect schema changes, missing records, duplicates, invalid values, and reconciliation issues, improving the reliability of downstream datasets for analytics and reporting.
- Implemented cloud infrastructure and DevOps practices using Terraform, Docker, GitHub Actions, AWS CodePipeline, AWS CodeBuild, AWS CodeDeploy, and Amazon EKS, automating infrastructure provisioning, application builds, testing, deployments, environment promotion, and controlled production releases.
- Secured and monitored production data platforms using AWS IAM, AWS Secrets Manager, Amazon CloudWatch, S3 security controls, metrics, dashboards, alarms, and centralized logging, applying least-privilege access and proactive monitoring to improve pipeline observability and operational reliability.
- Partnered with onshore and offshore data engineers, cloud engineers, application teams, and business stakeholders to analyze source systems, assess analytics use cases, integrate data, design scalable solutions, translate business requirements into technical solutions, and troubleshoot production issues across development, test, and production environments.
- Integrated cloud-native services, including AWS Lambda, SQS, SNS, EventBridge, REST APIs, Snowpark, and AWS Step Functions, to enable event-driven data ingestion and distributed data workflows while strengthening data governance, cataloging, technical documentation, and reusable engineering standards for more consistent development and production delivery.

Indegene **Jun 2023 - Jul 2024**

Data Engineer *Bengaluru | India*

- Designed, developed, and optimized scalable ETL and ELT pipelines using Dataiku, Python, SQL, and Snowflake to transform complex datasets into reliable, analytics-ready data products that supported business reporting and daily data operations.
- Wrote and optimized advanced Snowflake SQL queries, including complex joins, CTEs, window functions, aggregations, subqueries, and analytical transformations, improving query performance and supporting data analysis, reporting, and downstream business intelligence workloads.
- Built automated data processing and validation workflows using Python, Pandas, NumPy, Dataiku, and SQL, applying data quality checks, reconciliation rules, exception handling, and monitoring to detect data issues, support corrective actions, and improve pipeline reliability.
- Established and maintained data governance frameworks, data quality standards, metadata management practices, validation procedures, and operational controls, partnering with Data and Analytics Leads, business units, IT, and offshore teams to translate business requirements into reliable technical solutions and improve data accuracy, consistency, and usability.
- Managed data engineering delivery across Agile and DevOps environments by coordinating data analysts and engineers, tracking SLAs, milestones, risks, and dependencies, maintaining status reports, and collaborating with onshore and offshore stakeholders to manage delivery and resolve project risks.
- Improved ETL processes, Snowflake workloads, Dataiku flows, and operational procedures through data profiling, validation, and process testing, reducing processing time and operational risk while supporting dashboard and reporting teams with reliable datasets, decision-ready analysis, governance updates, and executive project status reports.

Precision Medicine Group **Oct 2022 - Jun 2023**

Data Analyst *Pune | India*

- Analyzed client operational and business datasets using SQL and Python (Pandas, NumPy) to identify data trends, discrepancies, operational issues, and process exceptions; translated findings into clear analytical outputs and recommendations for internal and client-facing stakeholders.
- Executed defined operational processes and analytical workflows by applying data validation, quality checks, reconciliation, and exception analysis across recurring deliverables, investigating discrepancies and documenting results to maintain accuracy, consistency, and adherence to established SOPs and governance requirements.
- Developed and enhanced Python-based data preparation and reporting workflows using Pandas, NumPy, and Jupyter, automating repetitive data-cleaning, transformation, validation, and reporting activities while improving turnaround time and reducing manual processing effort.
- Collaborated with Senior Data Analysts, Operations Delivery Leads, engineering teams, and cross-functional stakeholders across time zones to investigate client questions, test new or modified operational procedures, validate process changes, and recommend improvements that increased workflow efficiency and delivery quality.
- Maintained SOPs, process documentation, data definitions, validation procedures, testing results, and operational metrics while coordinating multiple client tasks, delivering analytical results within established timelines, and conducting secondary research and root-cause analysis to support continuous process improvement and client delivery.

PROJECT

Plant Disease Detection - Machine Learning Project **2025**

Northwest Missouri State University

- Built a plant disease classification solution using Python, TensorFlow, OpenCV, Pandas, and NumPy, performing image collection, dataset organization, preprocessing, feature engineering, and supervised model development across multiple plant disease categories.
- Developed and trained convolutional neural network models, tuning hyperparameters and comparing baseline and optimized architectures to improve classification accuracy, robustness, and overall model performance.
- Applied image augmentation techniques including rotation, flipping, and scaling to expand training data diversity, reduce overfitting, and strengthen model generalization across disease categories.
- Built a reproducible preprocessing-to-evaluation pipeline and optimized data-processing workflows to streamline experimentation, improve processing efficiency, validate structured dataset expansion, and accelerate model iteration.
- Evaluated and analyzed model results using accuracy, loss curves, confusion matrices, and misclassified samples, identifying performance gaps and guiding preprocessing and model improvements; documented experimental results and performance comparisons across iterations.

EDUCATION

Northwest Missouri State University **Aug 2024 - Dec 2025**

Master of Science | Applied Computer Science *Maryville | MO*

GITAM University **Sep 2020 - May 2024**

Bachelor of Technology | Computer Science and Engineering *INDIA*

CERTIFICATION

- DATAIKU Core Designer Certificate